1 Strategies for Merging Earthquake Catalogs: A Literature Review

The construction of unified earthquake catalogs from multiple heterogeneous source networks is one of the central methodological challenges of observational seismology. The need arises at every level of the discipline: global hazard models require decades of stable, magnitude-homogeneous records spanning multiple agencies (Pagani et al., 2014); regional seismic-rate studies must combine specialist national networks with global bulletins to fill spatial coverage gaps (Woessner and Wiemer, 2005); historical and pre-instrumental periods demand careful integration of macroseismic and early-instrumental data (Grünthal, 1998; Rovida et al., 2016); and the growth of dense, heterogeneous strong- motion and nodal-sensor networks creates new challenges at sub-local scales (Share et al., 2022).

Despite the ubiquity of the problem, no single standard algorithmic pipeline has emerged across the community. Different groups have developed independent solutions—from the rule-based proximity approaches of the Northern California Earthquake Data Center (2024) and the ISC bulletin (Bondár and Storchak, 2011), to the graph-theoretic frameworks of Benz et al. (2019), to the physics-based nearest-neighbour discriminant of Tanaka et al. (2022)—each optimised for different data volumes, network configurations, and seismicity regimes. The goal of this review is to synthesise the methodological state of the art into a coherent framework that can inform automated merging implementations.

1.1 The Two-Step Decomposition

Any robust merging pipeline must separate two operations that are logically distinct but are frequently conflated in practice:

1. **Event association and deduplication** — determining which records across source catalogs describe the *same physical earthquake* and grouping them into equivalence classes.

2. **Parameter selection and homogenisation** — from each equivalence class, deriving a single best origin time, hypocentre, and magnitude, and expressing all magnitudes on a common scale.

The distinction matters because applying a magnitude conversion formula *before* the association step—for instance to make thresholds magnitude-scale-independent—introduces feedback loops: converted magnitudes may shift an event across an association threshold, altering which pairs are grouped, which in turn changes the sample from which conversion coefficients are estimated (Storchak et al., 2013; Grünthal et al., 2016). The correct order is always: associate first using raw reported values; homogenise second using the confirmed groups.

1.2 Historical Development and the Heterogeneous Network Problem

1.2.1 The pre-digital era: 1900–1970

Early global earthquake catalogs—the Gutenberg-Richter Seismicity of the Earth (Gutenberg and Richter, 1954), the Abe catalog of large historical earthquakes (Abe, 1981), and the pre-1964 portion of the Engdahl-van der Hilst-Buland (EHB) dataset (Engdahl et al., 1998)—were assembled by manual examination of seismogram archives, station bulletins, and published macroseismic accounts. Cross-network event identification was performed by inspection, using timing differences of several minutes and epicentral discrepancies of hundreds of kilometres as acceptable tolerances for the analogue recording era (Bondár and Storchak, 2011).

The International Seismological Summary (ISS, predecessor to the ISC bulletin) began assembling global phase data in 1918. Its practice of retaining all contributing agency solutions alongside a centralised re-location—still followed by the ISC today—established the principle of *solution provenance* that underpins modern best practice (Storchak et al., 2020).

1.2.2 The WWSSN and NEIC era: 1960–1990

The deployment of the World-Wide Standardized Seismograph Network (WWSSN) from 1961 onward provided the first globally uniform recording infrastructure and enabled systematic relocation of events at $M \geq 5$ using common velocity models (Engdahl et al., 1998). The 1964 establishment of the National Earthquake Information Center (NEIC, initially the National Earthquake Information Service) created the first dedicated fusion centre for global earthquake data, introducing the concept of a *preferred solution* from a ranked list of contributing agencies (USGS, 2025).

The ANSS Composite catalog, which consolidates the NEIC global bulletin with authoritative regional networks across North America, employs a proximity rule of 16 s in time and 100 km in distance to identify duplicate reports submitted by different networks for the same event (Northern California Earthquake Data Center, 2024). This rule was formulated for the digital WWSSN era and remains the most precisely documented operational threshold in any publicly available system documentation.

1.2.3 The digital broadband era: 1990–present

The transition to digital broadband recording in the late 1980s and 1990s, the expansion of the IRIS/IDA Global Seismographic Network (GSN), and the growth of the FDSN data-exchange standard (ETH Zürich Seismology / QuakeML Working Group, 2015) transformed the scale of the deduplication problem. By 2010, the ISC bulletin was receiving origin reports from more than 130 contributing agencies worldwide, each applying different velocity models, phase identification algorithms, and location programs (Storchak et al., 2020). The ISC-GEM Global Instrumental Earthquake Catalog (Storchak et al., 2013, 2018), released in 2013 and subsequently extended through 2021, represents the most comprehensive attempt to retrospectively homogenise this heterogeneous archive for events $M \geq 5.5$.

The proliferation of dense regional networks—GeoNet in Aotearoa New Zealand (GeoNet, 2025a), Hi-net in Japan (Obara et al., 2005), the Southern California Seismic Network (Hutton et al., 2010), and the European EPOS infrastructure (EPOS, 2025)—has simultaneously improved detection thresholds for small events ($M < 3$) while creating new challenges for merging: specialist local networks now routinely detect thousands of microseismic events per year that global agencies never report, making one-to-one association across scales difficult (Woessner and Wiemer, 2005; Illsley-Kemp and Mestel, 2025).

1.3 Event Association: Spatio-Temporal Matching Algorithms

1.3.1 Physical basis for proximity thresholds

The rationale for any spatio-temporal matching window is that the reported origin time and hypocentre of two records for the same physical earthquake will differ only by the combined effect of (i) timing errors at each recording network, (ii) location uncertainties arising from network geometry and velocity model errors, and (iii) any deliberate choices such as depth fixing.

For digital networks with GPS-synchronised clocks, origin time errors are typically well below 1 s (Bondár and Storchak, 2011). Location errors depend strongly on network geometry: the azimuthal gap (the largest angular gap between contributing stations) and the minimum station distance are the dominant predictors. Bondár et al. (2004) showed empirically from the ISC bulletin that the 90th percentile of location error is approximately:

$$\sigma_r \approx 0.51 + 0.038\,\mathrm{GAP} + 0.0055\,\Delta_{\min}^{-1}, \tag{1}$$

where GAP is the azimuthal gap in degrees and $\Delta_{\min}$ is the closest station distance in degrees. For a typical global network with GAP $\approx 120°$ and $\Delta_{\min} \approx 3°$, this yields $\sigma_r \approx 5$ km. For poorly-constrained events with GAP $\approx 270°$, σ_r can exceed 50 km, justifying wider matching windows.

Pre-digital era timing errors of several seconds per station translate to origin time uncertainties of 1–10 s for well-recorded events and up to 30–60 s for poorly-recorded events in the pre-1960 period (Bondár and Storchak, 2011). This is the physical basis for the wider windows in table 1.

1.3.2 Window-based forward-sweep

The oldest and most widely deployed method sorts all incoming origin reports by origin time and scans forward with a sliding window. If two reports from *different* networks fall within a fixed time window $\Delta t_{\max}$ and an epicentral distance $\Delta d_{\max}$, they are flagged as candidate duplicates (Northern California Earthquake Data Center, 2024; Woessner and Wiemer, 2005). Table 1 summarises documented threshold choices from major operational systems.

Table 1: Spatio-temporal matching thresholds documented in major operational earthquake catalog systems. Δt = origin-time window, Δd = epicentral distance window. ANSS thresholds apply to inter-network duplicate pairs only; intra-network deduplication is assumed to have been handled upstream by each contributing agency.

System / Study	Δt (s)	Δd (km)	Reference
ANSS Composite catalog	16	100	Northern California Earthquake Data Center (2024)
Typical global build ($M \geq 4.5$)	60	100	Storchak et al. (2013)
Local events ($M < 4$)	30	30	Woessner and Wiemer (2005)
Pre-digital era (pre-1960)	300	200	Bondár and Storchak (2011)
Dead Sea Transform region	60	100	Sira et al. (2022)
Mexico unified catalog	60	≈ 111 (1°)	Zúñiga et al. (2019)
Turkey homogeneous catalog	60	100	Tan (2021)
Italy CPTI15 pipeline	30–120	50–150	Rovida et al. (2016)
ISC-GEM global ($M \geq 5.5$)	60	100	Storchak et al. (2018)

The 16 s / 100 km rule of the ANSS is the most precisely documented publicly available threshold for a production system. It was chosen empirically to balance false-positive (erroneously merging distinct earthquakes) against false-negative (missing genuine duplicate pairs) rates for the population of events in the North American digital-era database (Northern California Earthquake Data Center, 2024). For teleseismic events the 60 s / 100 km window is widely cited; the ISC uses a comparable criterion internally when auto-grouping agency solutions prior to relocation (Storchak et al., 2020).

Algorithmic complexity and spatial indexing. A naive implementation of the forward sweep compares all ordered pairs (E_i, E_j) with $j > i$, giving $O(n^2)$ complexity for n events. For large catalogs ($n > 10^5$), this is prohibitive. The standard optimisation builds a *spatial index* over event locations: a uniform grid (hash grid), k-d tree, or R-tree, allowing events within $\Delta d_{\max}$ to be retrieved in $O(\log n)$ time. Combined with a time-sorted event list and an early-exit rule when the candidate's origin time exceeds $t_i + \Delta t_{\max}$, the algorithm achieves $O(n \log n)$ overall complexity in the typical case (Benz et al., 2019).

Grid-based indexes use a fixed cell size calibrated to the distance threshold. At high latitudes, degrees of longitude subtend shorter distances, so the cell size must be adjusted by $\cos(\phi)$ where ϕ is the latitude (Tan, 2021). Near the International Date Line (longitude ±180°), grid cell keys must wrap correctly; failure to implement wrapping silently loses candidates for events in the Pacific (Sira et al., 2022).

Adaptive thresholds. Fixed windows perform poorly across the full magnitude range because larger events are subject to greater timing and location uncertainty at distant networks. The standard remedy, used in the ISC-GEM pipeline, is to scale the effective window by event magnitude and depth (Storchak et al., 2018):

$$\Delta t_{\text{eff}} = \Delta t_{\text{config}} \cdot f_t(M), \qquad \Delta d_{\text{eff}} = \Delta d_{\text{config}} \cdot f_d(M) \cdot g(h), \tag{2}$$

where representative multiplier functions consistent with ISC-GEM practice (Storchak et al., 2018; Engdahl et al., 2020) are:

$$f_t(M) = \begin{cases} 1.0 & M < 4.0 \\ 1.5 & 4.0 \le M < 5.5 \\ 2.0 & 5.5 \le M < 7.0 \\ 3.0 & M \ge 7.0 \end{cases} \tag{3}$$

$$f_d(M) = \begin{cases} 1.0 & M < 4.0 \\ 1.5 & 4.0 \le M < 5.5 \\ 2.5 & 5.5 \le M < 7.0 \\ 4.0 & M \ge 7.0 \end{cases} \tag{4}$$

$$g(h) = \begin{cases} 1.0 & h < 100\,\text{km} \\ 1.2 & 100 \le h < 300\,\text{km} \\ 1.5 & h \ge 300\,\text{km} \end{cases} \tag{5}$$

The depth multiplier $g(h)$ reflects the well-established result that intermediate and deep-focus earthquakes are harder to locate precisely because depth-sensitive phases (pP, sP) are often absent from or misidentified in network phase datasets (Engdahl et al., 1998, 2020). For $M \ge 7$ deep events, the combined multiplier can reach $4.0 \times 1.5 = 6.0$, requiring the spatial search radius to extend to six times the configured baseline threshold.

Early termination and threshold consistency. A subtle but important correctness issue arises when combining the forward-sweep time-sorted early exit with adaptive thresholds. The early-exit condition

$$t_j - t_i > \Delta t_{\text{config}} \cdot f_t(M_i) \tag{6}$$

uses only the anchor event's magnitude M_i. However, the adaptive matching function (2) uses the *average* of both event magnitudes. If the candidate j has a substantially higher magnitude than i, the correct effective threshold (evaluated at the average magnitude) may exceed the early-exit threshold (evaluated at M_i alone), causing valid matches to be discarded. The conservative fix is to replace $f_t(M_i)$ in (6) with $f_t(\max(M_i, M_j))$, or equivalently to use the maximum expected candidate magnitude (e.g. $M_i + 2$) as a conservative upper bound when evaluating the exit condition (Benz et al., 2019).

Limitation: dense aftershock sequences. During intense aftershock sequences the inter-event time between consecutive legitimate aftershocks can fall well below the duplicate-detection window. Tanaka et al. (2022) demonstrated that in the 24 hours following the 2011 Tōhoku $M_{\text{w}}\,9.0$, more than 599 genuine aftershock pairs had inter-event times under 60 s — the same interval used as a duplicate window in the ISC-GEM pipeline. Fixed-window methods therefore cannot discriminate a duplicate pair from an aftershock pair in high-seismicity periods without additional physical information, a problem that grows more severe as network density increases and detection completeness lowers the magnitude threshold into the range where aftershock rates are high (Hainzl et al., 2016).

1.3.3 Nearest-neighbour proximity function

Tanaka et al. (2022) introduced a physics-based proximity function that discriminates duplicate pairs from aftershock pairs by exploiting their different topological signatures in event-pair space. The method builds on the nearest-neighbour statistics introduced by Zaliapin et al. (2008) for aftershock sequence identification. Genuine duplicates form *isolated pairs* in rescaled event space — their nearest-neighbour distances cluster tightly below a network-error-derived threshold — whereas aftershocks form *branching trees* following Omori-law temporal decay (Omori, 1894; Utsu, 1961).

The proximity between two events i and j (with j following i in time) is defined as

$$\eta(i, j) = t_{ij}\, r_{ij}^{d_f}\, 10^{-bm_i}, \tag{7}$$

where t_{ij} is the inter-event time (s), r_{ij} is the epicentral distance (km), $d_f \approx 1.6$ is the fractal dimension of the epicentre distribution (Kagan, 1991), $b \approx 1.0$ is the Gutenberg-Richter b-value (Aki, 1965), and m_i is the magnitude of the preceding event. In this rescaled space, aftershocks cluster near their parent event (small η), while duplicates cluster near zero regardless of the parent-event magnitude (Zaliapin et al., 2008; Tanaka et al., 2022).

The duplicate-detection threshold in η-space is calibrated against the distribution of location errors for the specific network under analysis, not against earthquake physics. This is the key advantage: the threshold adapts to network capability rather than assuming a fixed proximity tolerance (Tanaka et al., 2022). Applied to the Tōhoku sequence, the method achieved duplicate identification reliability $> 97\,\%$ even during periods with aftershock rates of several hundred events per hour.

Parameter estimation. The fractal dimension d_f can be estimated from the slope of the correlation integral of hypocentre coordinates in log-log space (Kagan, 1991). The b-value is estimated via maximum likelihood from the observed frequency-magnitude distribution (Aki, 1965), with magnitude of completeness M_c determined by the Wiemer and Wyss (2000) goodness-of-fit method before fitting. Both estimates should be computed from the target region's seismicity, not borrowed from global averages, as both parameters show significant regional variability (Woessner and Wiemer, 2005).

Limitations. The nearest-neighbour method performs less well for very sparse catalogs where the nearest-neighbour distance distribution is dominated by background seismicity rather than aftershock clusters, reducing the bimodal separation that enables discrimination (Hainzl et al., 2016). It also requires a minimum catalog length (typically several years) to estimate b and d_f reliably.

1.3.4 Graph-based connected-components association

The methodologically strongest general-purpose approach constructs an undirected association graph $G = (V, E)$ in which each node $v \in V$ is an origin report and each edge $e \in E$ connects a pair of reports that satisfy a proximity criterion (Benz et al., 2019; Zaliapin and Ben-Zion, 2013). Connected components of G, recoverable in $O(n\,\alpha(n))$ time via the Union-Find (Disjoint-Set Union) algorithm (Tarjan, 1975) — where α is the inverse Ackermann function and is effectively constant for all practical n — correspond to clusters of reports describing the same physical earthquake.

Algorithm.

1. Sort all origin reports by origin time; build a spatial index (uniform grid, k-d tree, or R-tree) over event hypocentres.
2. For each report E_i, query the index for all reports within $\Delta t_{\max}$ and $\Delta d_{\max}$ from networks *other than* the network of E_i.
3. For each qualifying candidate pair (E_i, E_j), add edge e_{ij} to G.
4. Find all connected components of G using Union-Find.
5. For each component, apply parameter-selection rules (section 1.9) to derive a single merged event record.

Advantage over the greedy forward sweep. The greedy sweep implicitly assumes that pairwise matching is transitive: if event A matches B, and B matches C, all three are merged. In the sweep implementation this is only guaranteed when events are processed strictly in time order and no candidate is marked processed before its later matches are evaluated. When the sweep marks B as processed upon matching with A, a subsequent evaluation of C against B is never performed, and the A-B-C cluster is incorrectly split (Benz et al., 2019). The graph approach has no such pathology: transitivity is guaranteed by the connected-component structure.

Integer linear programming refinement. Benz et al. (2019) extended the graph approach by applying Integer Linear Programming (ILP) to resolve ambiguous components — cases where a connected component may span two closely spaced but genuinely distinct earthquakes. The ILP minimises the total number of source events k required to explain all edges in a component, subject to the constraint that no two reports assigned to the same source event originate from the same network:

$$\text{minimise} \quad k \quad \text{subject to} \quad \forall (i,j) \in E : \text{source}(i) = \text{source}(j), \quad \forall i,j : \text{net}(i) = \text{net}(j) \Rightarrow \text{source}(i) \neq \text{sourc} \tag{8}$$

Applied to the Chilean catalogue 2010–2017, this two-stage approach detected 817 000+ earthquakes at $> 95\,\%$ accuracy against a synthetic validation dataset generated by forward modelling known network detection probabilities.

The ILP constraint that *no two reports from the same network belong to the same source event* is the algorithmic expression of what we term the *same-agency duplicate flag*: if two records from the same agency fall within the proximity window, they almost certainly describe two distinct physical earthquakes, not two observations of one (Benz et al., 2019; Tanaka et al., 2022).

The USGS NEIC Hydra production system uses a graph-association framework conceptually equivalent to the first stage of Benz et al. (2019) (USGS, 2025).

1.4 Location Uncertainty and Its Effect on Matching

1.4.1 Sources of location uncertainty

Hypocentre location uncertainty arises from five primary sources (Bondár et al., 2004; Engdahl et al., 1998):

1. **Phase picking errors.** Manual or automated picks of P, S, and depth phases carry random timing errors (typically 0.1–1 s for digital data, 1–5 s for analogue).

2. **Network geometry.** The azimuthal gap and minimum station distance control the conditioning of the location inversion. A gap $> 180°$ leaves the hypocentre poorly constrained in the ungapped direction; a large minimum distance makes depth poorly constrained.

3. **Velocity model errors.** All location algorithms use an approximation to Earth structure; residual travel-time prediction errors are correlated among nearby stations, biasing the location estimate. Bondár and Storchak (2011) introduced a correlated error covariance model to account for this.

4. **Depth-phase absence.** Without pP, sP, or pwP phases, depth is often fixed at a default value (10, 33, or 100 km depending on agency convention), introducing a systematic depth error that can exceed 50 km (Engdahl et al., 1998, 2020).

5. **Reporting delays and partial phase sets.** Rapid preliminary solutions from real-time networks (e.g. NEIC PDL) use fewer phases than the reviewed bulletin solution and therefore carry larger uncertainties (USGS, 2025).

1.4.2 Quantifying location quality for merging decisions

The ISC bulletin quality scoring system (Bondár and Storchak, 2011) provides a practical framework for weighting competing hypocentre solutions within a merged group. The primary quality indicators are:

- Azimuthal gap: GAP $< 180°$ for acceptable, $< 120°$ for good.

- Secondary azimuthal gap (gap ignoring the single most distant station): $< 180°$ is the ISC-GEM criterion for a usable depth (Storchak et al., 2018).

- Number of defining phases: ≥ 6 minimum; ≥ 20 for well- constrained depth.

- RMS travel-time residual: < 1.0 s for well-located events (Bondár and Storchak, 2011); the ISC bulletin flags events with RMS > 2.0 s as poorly located (Storchak et al., 2020).

- Horizontal uncertainty: expressed as the semi-major axis of the 90 % confidence ellipse in km (Bondár et al., 2004).

These metrics should be used to rank competing solutions within a merged event group (section 1.9), not merely to flag events after the fact.

1.4.3 Depth fixing and its consequences

A significant fraction of agencies fix the focal depth at a standard value when insufficient depth-sensitive phases are available. Engdahl et al. (2020) estimate that approximately 30 % of events in the pre-1990 ISC bulletin have fixed depths. Fixed-depth solutions carry no depth uncertainty information and should be assigned lowest priority in any depth-selection step. The depth type (free or fixed, and if fixed, the value and reason) should be preserved as event metadata to enable downstream filtering (Storchak et al., 2018; Engdahl et al., 2020).

The three most common fixed-depth conventions are 10 km (shallow crustal default, used by many regional agencies), 33 km (Conrad discontinuity, a historical NEIC convention), and 100 km (subduction-zone intermediate-depth default). Events with these three depths should be treated with heightened scrutiny in any depth-sensitive analysis (Engdahl et al., 1998).

1.5 The ISC-GEM Reference Methodology

The ISC-GEM Global Instrumental Earthquake Catalog (Storchak et al., 2013, 2018) is the most thoroughly documented end-to-end merging pipeline in the seismological literature. Its fifth revision (Storchak et al., 2018) covers 1904–2014 for $M_\mathrm{w} \geq 5.5$ globally and $M_\mathrm{w} \geq 5.0$ over continental regions, with subsequent extensions to 2021. The pipeline is described in sufficient detail to serve as a reference implementation for any automated merging system.

1.5.1 Data sources and ingestion

Source data include the ISC bulletin (phase data and agency origin solutions from 1904 onward), the NEIC PDE and predecessors, GCMT solutions from 1976 (Dziewonski et al., 1981; Ekström et al., 2012), regional CMT and MT catalogues (GeoNet, INGV, GEOFON, ETH, NIED), and macroseismic intensity data for the pre-1960 period (Storchak et al., 2013). Each source is assigned a priority tier and a data quality flag based on known limitations (e.g. analogue vs. digital recording, reporting latency, known velocity model biases).

1.5.2 Event association

The ISC automated grouping algorithm compares origin times and epicentre coordinates from all contributing agencies and assembles candidate groups using a proximity-scoring system that weights azimuthal gap, phase count, distance range, and network geometry (Bondár and Storchak, 2011). Groups that appear to span two distinct earthquakes — identified by anomalously large within-group hypocentre scatter or by the same-agency rule — are split by the ISC analyst team (Storchak et al., 2020).

1.5.3 Relocation using all available phase data

The ISC does not simply select the best existing agency solution; it *re-locates every event* from scratch using all contributed phase data. The two-stage relocation procedure is:

1. **EHB algorithm** (Engdahl et al., 1998): an iterative relocation using the ak135 reference velocity model (Kennett et al., 1995) with improved phase identification, particularly for pP and sP depth phases. The EHB procedure reduced systematic depth biases in the pre-1990 bulletin by 10–30 km for subduction-zone events (Engdahl et al., 1998).

2. **ISC location algorithm** (Bondár and Storchak, 2011): fixes the depth output from EHB and further reduces systematic hypocentre bias by modelling the spatial correlation of travel-time prediction errors using an empirical covariance function estimated from the bulletin's residual distribution.

The ISC-authored solution is designated *prime* when at least 6 defining phases are available and the azimuthal gap is $< 180°$. Where these criteria are not met, the highest-ranked contributing agency solution is used as prime (Storchak et al., 2018).

Crucially, *all* agency solutions are retained as secondary (non-prime) hypocentres in the ISC bulletin. This traceability model — derived from the practice of the original ISS — ensures that all downstream applications can reproduce the selection logic and compare solutions across agencies (Storchak et al., 2020).

1.5.4 Magnitude assembly in ISC-GEM

ISC-GEM reports a single M_w per event using the following protocol (Storchak et al., 2018):

1. **Direct** M_w from global moment-tensor catalogs: GCMT (Dziewonski et al., 1981; Ekström et al., 2012) (preferred), USGS W-phase, EMSC MT, GeoNet CMT, ETH CMT.

2. M_w **proxy from** M_s: the ISC re-computes M_s using an *alpha-trimmed median* (10 % trimming on each tail) of individual station amplitudes, suppressing outliers from poorly calibrated stations. The resulting M_s is converted to M_w via an exponential (non-linear) regression.

3. M_w **proxy from** m_b: similarly trimmed and converted via a non-linear regression that accounts for m_b saturation above $\approx M_\mathrm{w}\,6.2$.

The alpha-trimmed median is a robust location estimator in the statistical sense: its breakdown point (the fraction of outliers that can corrupt the estimate) is 10 %, compared with 0 % for the simple mean (Huber, 1981). For the station-level amplitude data used to compute M_s, where a single poorly calibrated station can deviate by 1–2 magnitude units, this robustness is practically important.

1.6 Magnitude Homogenisation: Theory and Practice

1.6.1 The physical basis for moment magnitude

Moment magnitude M_w was introduced by Hanks and Kanamori (1979) as a logarithmic measure of the seismic moment M_0 (measured in dyne-cm):

$$M_\mathrm{w} = \tfrac{2}{3}(\log_{10} M_0 - 16.1). \tag{9}$$

The seismic moment $M_0 = \mu A\bar{D}$, where μ is the shear modulus, A is the fault rupture area, and $\bar{D}$ is the mean slip, is a physically meaningful measure of earthquake size that does not saturate at large magnitudes (Kanamori, 1977). In contrast, m_b saturates above $M_\mathrm{w} \approx 6.0$–6.2 (because short-period body waves become insensitive to the growing fault area), and M_s saturates above $M_\mathrm{w} \approx 8.0$–8.2 (Bormann, 2012).

The saturation problem directly affects merging: a network reporting $m_\mathrm{b} = 6.5$ for an M_w 7.5 event is not recording a different earthquake; it is simply using a saturated scale. Magnitude-based filtering or association thresholds that do not account for scale-dependent saturation will therefore misclassify or incorrectly reject legitimate matches (Bormann, 2012; Storchak et al., 2018).

1.6.2 Regression method: orthogonal vs. ordinary least squares

The fundamental flaw in using Ordinary Least Squares (OLS) for magnitude conversion is that OLS minimises the sum of squared vertical residuals, implicitly assuming the predictor variable is measured without error. When both the predictor M_x and the response M_w carry measurement error of comparable magnitude, OLS produces a biased slope estimator (Fuller, 1987):

$$\hat{\beta}_\mathrm{OLS} = \beta \cdot \frac{\sigma^2_{M_x}}{\sigma^2_{M_x} + \sigma^2_\varepsilon}, \tag{10}$$

where β is the true slope, $\sigma^2_{M_x}$ is the variance of the true predictor, and σ^2_ε is the measurement error variance. The ratio is always ≤ 1, so OLS systematically underestimates the true conversion slope — an *attenuation bias* that compresses the converted magnitude range and biases b-value estimates from the homogenised catalog (Grünthal et al., 2016; Carroll et al., 2006).

Generalised Orthogonal Regression. The solution is **Generalised Orthogonal Regression** (GOR), which minimises the sum of squared perpendicular distances weighted by the ratio $\eta = \sigma^2_{M_\mathrm{w}}/\sigma^2_{M_x}$ (Grünthal et al., 2016; Akkar et al., 2014; Cotton et al., 2006). The GOR slope is:

$$\hat{\beta}_\mathrm{GOR} = \frac{(SYY - \eta\, SXX) + \sqrt{(SYY - \eta\, SXX)^2 + 4\eta\, SXY^2}}{2\, SXY}, \tag{11}$$

where SXX, SYY, SXY are the corrected sums of squares and cross-products. When $\eta = 1$ (equal uncertainties), GOR reduces to the standard geometric mean (GM) regression; when $\eta \to 0$ (predictor error-free), GOR approaches OLS; and when $\eta \to \infty$ (response error-free), GOR approaches the reverse OLS regression of M_x on M_w (Fuller, 1987).

The SHEEC (SHARE European Earthquake Catalogue) project (Grünthal et al., 2016) provides published GOR coefficients for M_{L}, M_{s}, and m_{b} to M_{w} conversion for European seismicity, estimated from the overlap period of the EMS98 national catalogs and the ISC-GEM dataset. These coefficients are recommended as defaults for European and Mediterranean merging applications (Grünthal et al., 2016).

Non-linear functional forms. For m_{b} and M_{s}, the ISC-GEM pipeline (Storchak et al., 2018) chose exponential (non-linear) conversion models rather than linear GOR, on the basis that the physical saturation of both scales introduces a non-linear relationship at the extremes that linear regression cannot capture. Piecewise-linear models with optimised crossover points — as implemented in the CPTI15 Italian catalog pipeline using the Gasperini et al. (2012) framework — offer a practical middle ground that approximates non-linearity without the extrapolation risks of exponential models.

The Scordilis (2006) coefficients. The widely used Scordilis (2006) linear relationships (e.g. $M_{\mathrm{w}} = 0.67\,M_{\mathrm{L}} + 1.17$, $M_{\mathrm{w}} = 0.85\,m_{\mathrm{b}} + 1.03$, $M_{\mathrm{w}} = 0.67\,M_{\mathrm{s}} + 2.07$ for $M_{\mathrm{s}} < 6.2$) are OLS regressions on a global dataset. As such they carry the attenuation bias in (10) and are known to systematically underestimate M_{w} for large events near each scale's saturation boundary (Grünthal et al., 2016). They are appropriate as a first approximation and for applications where the bias is small compared to other uncertainties, but should be replaced by GOR-derived relationships for research-grade catalogs.

1.6.3 Magnitude type priority hierarchy

The ISC-GEM priority order (Storchak et al., 2018), consistent with Grünthal et al. (2016) and the ISC bulletin operational practice, is:

1. M_{w} from GCMT waveform inversion (Dziewonski et al., 1981; Ekström et al., 2012) — gold standard; does not require conversion.
2. M_{w} from USGS/NEIC W-phase or broadband CMT solution.
3. M_{w} from regional CMT (GeoNet (GeoNet, 2025a), INGV, GEOFON/GFZ, ETH).
4. M_{w} proxy from M_{s} (preferred over m_{b} because M_{s} saturates at higher magnitude and the conversion scatter is smaller (Bormann, 2012)).
5. M_{w} proxy from m_{b}.
6. M_{w} proxy from M_{L} (most region-dependent; highly variable station corrections).
7. M_{w} proxy from M_{d} or M_{c} (duration / coda magnitude; least reliable, used only when nothing else is available).

Within each priority tier, the estimate with lower reported uncertainty is preferred (Storchak et al., 2018). Where no uncertainty is reported, network authority from table 2 acts as a tie-breaker.

1.6.4 Bayesian magnitude merging across networks

Simple priority selection discards valid information when multiple authoritative M_w estimates exist for the same event. Si et al. (2024) proposed a Bayesian framework in which each network i's observed magnitude is modelled as a linear function of the unknown true magnitude M_true:

$$M_{\mathrm{obs},i} = a_i + b_i\, M_\mathrm{true} + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0,\, \sigma_i^2), \tag{12}$$

where a_i and b_i are network-specific intercept and slope (estimated from the overlap period), and σ_i^2 is the within-network reporting variance. A Gutenberg-Richter prior $p(M_\mathrm{true}) \propto 10^{-bM_\mathrm{true}}$ is placed on the true magnitude. The posterior mean is:

$$\hat{M}_\mathrm{true} = \frac{\sum_i b_i(M_{\mathrm{obs},i} - a_i)/\sigma_i^2 - b\ln(10)\cdot\sigma^2_\mathrm{post}}{\sum_i b_i^2/\sigma_i^2}, \tag{13}$$

where $\sigma^2_\mathrm{post} = (\sum_i b_i^2/\sigma_i^2)^{-1}$ is the posterior variance. When two authoritative solutions agree within $0.1\, M_\mathrm{w}$ units, the posterior is insensitive to the choice of b; when they disagree by $> 0.3\, M_\mathrm{w}$ units, a data quality flag should be raised regardless of which method is used (Si et al., 2024).

Uncertainty propagation for converted magnitudes. When M_w is estimated via conversion from M_x, the total uncertainty is (Sira et al., 2022; Grünthal et al., 2016):

$$\sigma^2_{M_\mathrm{w}} = \sigma^2_{\hat{\alpha}} + \left(\frac{\partial f}{\partial M_x}\right)^2 \sigma^2_{M_x} + \sigma^2_\mathrm{model}, \tag{14}$$

where $\hat{\alpha}$ is the conversion-equation intercept (with uncertainty $\sigma^2_{\hat{\alpha}}$ from the GOR fit), $f(M_x)$ is the conversion function, $\sigma^2_{M_x}$ is the individual measurement uncertainty of the predictor magnitude, and σ^2_model is the residual scatter of the regression (model uncertainty). For the Scordilis $m_\mathrm{b} \to M_\mathrm{w}$ conversion with slope ≈ 0.85, the conversion amplifies the input uncertainty by a factor $0.85^2 \approx 0.72$ (a reduction), but the model residual ($\sigma_\mathrm{model} \approx 0.3$) typically dominates (Scordilis, 2006; Grünthal et al., 2016).

1.6.5 Combining macroseismic and instrumental estimates

Where both macroseismic ($M_{\mathrm{w,macro}}$, derived from intensity data via Grünthal 1998) and instrumental ($M_{\mathrm{w,inst}}$) estimates exist—a common situation for the historical period 1900–1960—the CPTI15 Italian pipeline (Rovida et al., 2016) uses inverse-variance weighting:

$$M_{\mathrm{w,comb}} = \frac{M_{\mathrm{w,macro}}/\sigma^2_\mathrm{macro} + M_{\mathrm{w,inst}}/\sigma^2_\mathrm{inst}}{1/\sigma^2_\mathrm{macro} + 1/\sigma^2_\mathrm{inst}}. \tag{15}$$

This is the minimum-variance unbiased estimator under the assumption of independent Gaussian errors (Kay, 1993). For the Italian catalog, $\sigma_\mathrm{macro} \approx 0.3$–$0.5$ and $\sigma_\mathrm{inst} \approx 0.1$–$0.2$ (Rovida et al., 2016), so the combined estimate is dominated by the instrumental solution when both are available but gracefully falls back to the macroseismic estimate for events with no instrumental record.

1.7 Network Authority Hierarchies and Regional Overrides

1.7.1 The philosophical distinction: selection vs. re-analysis

Two fundamentally different philosophies underlie network authority models:

Selection model (ANSS ComCat). Each contributing network submits its preferred solution. A weight is assigned to each network per product type; the highest-weight solution is the "preferred" product (USGS, 2025). Where weights are equal, the most recently updated solution takes precedence. No re-analysis is performed by the aggregator.

Re-analysis model (ISC). The aggregator receives all raw phase data and performs its own relocation and magnitude computation using all contributed data simultaneously. The aggregator's solution is prime; contributing solutions are retained as secondary (Storchak et al., 2020).

For research-grade applications, the re-analysis model is strongly preferred because it provides a single uniform solution computed with a common algorithm and velocity model, eliminating inter-agency biases (Storchak et al., 2013). For rapid operational applications where relocation latency is unacceptable, the selection model is appropriate.

1.7.2 Regional authority and the "authoritative network" principle

The ANSS ComCat implements the *authoritative network* principle: for earthquakes occurring within a defined geographic footprint, the regional specialist network is given the highest preferred weight, overriding the NEIC global solution (USGS, 2025). Examples include the Caltech/USGS Southern California Seismic Network (SCSN, code `ci`) for Southern California, the University of Utah Seismograph Stations (`uu`) for the Intermountain West, and GeoNet (`nz`) for Aotearoa New Zealand (GeoNet, 2025a).

The justification is that regional networks operate denser station arrays with local velocity models specifically calibrated to regional structure, producing lower-uncertainty locations and completeness to smaller magnitudes than any global agency (Woessner and Wiemer, 2005; Hutton et al., 2010). For the New Zealand region, GeoNet (2025a) document that GeoNet solutions achieve completeness to approximately $M_{\mathrm{L}}\,1.5$ in the North Island, compared to the ISC's practical completeness limit of $\approx M\,4.5$ for the same region.

Table 2: Recommended global network authority hierarchy for parameter selection. Priority 1 = highest authority. Regional overrides (marked with †) take precedence over all entries above them in the table for earthquakes within the network's geographic footprint.

Priority	Network / Agency	Code(s)	Notes
1	GCMT	`gcmt`	Waveform CMT; gold-standard M_w
2	ISC / ISC-GEM	`isc`	Re-located; full phase data retained
3	USGS NEIC	`us`	W-phase M_w; global coverage
4	EMSC	`emsc`	Euro-Med aggregator
5	GEOFON / GFZ	`gfz`	Broadband; rapid solutions
6†	GeoNet / GNS	`nz`	Regional override: Aotearoa NZ
6†	JMA	`jp`	Regional override: Japan
6†	INGV	`ingv`	Regional override: Italy
6†	IGN	`ign`	Regional override: Spain
6†	BGR / GFZ	`de`	Regional override: Germany
7	IRIS / FDSN	`ir`	Waveform archive; no preferred solution

1.7.3 Temporal variation in network authority

Network capabilities change over time: networks expand or contract, recording equipment is upgraded, velocity models are revised. The authority hierarchy should therefore be applied with awareness of the recording era. For example, the GCMT catalog begins in 1976 (Dziewonski et al., 1981); for events before that date, the tier-1 authority falls to ISC or NEIC. GeoNet's digital broadband network was not fully operational until the early 2000s; for pre-2000 New Zealand events, ISC or the GNS analogue bulletin should be used as the primary source (GeoNet, 2025a).

1.8 Regional Catalogue Merging: Case Studies

1.8.1 Dead Sea Transform region

Sira et al. (2022) described an automated pipeline for the Dead Sea Transform region (DST-CAT), covering 1900–2017 and $M_w \geq 3.0$, assembled from ISC, the Geophysical Institute of Israel (GII), and several regional networks. Key methodological choices were:

- **Association**: 60 s / 100 km window with manual review of borderline cases.
- **Preferred solution**: specialist regional agency (GII) for events within the DST zone; ISC for the broader region; GCMT where available.

- **Magnitude conversion**: GOR-based M_L, M_s, and m_b to M_w relationships derived from the 1976–2017 overlap period, with uncertainties propagated via (14).
- **Era-dependent windows**: wider windows (90 s / 150 km) for the pre-1960 period to account for analogue timing errors.

The study explicitly noted the failure of fixed-window methods during the 1995 M_w 7.2 Aqaba mainshock sequence, where aftershock rates exceeded 10 events per hour for two days, and adopted a hybrid approach with manual review for the highest-rate periods.

1.8.2 Mexico unified catalogue

Zúñiga et al. (2019) assembled a unified Mexican seismicity catalog (1750–2018, $M \geq 3$) from seven source catalogs including the SSN (Servicio Sismológico Nacional), NEIC PDE, ISC, and the Reyes-Dávila regional bulletins. The deduplication procedure used a 60 s / 1° ($\approx$ 111 km) window and a source priority list with the SSN as authoritative for events within Mexico. The study documents a magnitude change point in 1963 (transition from analogue to digital recording) requiring separate completeness analyses for the two periods.

1.8.3 Turkey homogeneous catalogue

Tan (2021) constructed a homogeneous M_w catalog for Turkey (1900–2018) from TURKNET, AFAD, ISC, GCMT, and several historical sources. The study used a 60 s / 100 km proximity window and applied GOR conversions with era-specific coefficients. A key finding was that the Scordilis OLS coefficients produced a systematic overestimate of M_w for Turkish M_L values below 3.5, by approximately 0.2 units — confirming the OLS attenuation bias described in section 1.6.2. The homogenised catalog was then declustered using the Reasenberg (1985) algorithm before use in seismic hazard calculations.

1.8.4 Italy: CPTI15

The Parametric Catalogue of Italian Earthquakes (CPTI15, Rovida et al. 2016) is the most comprehensive regional merging effort for a seismically active European country, covering 1000–2014 with $M_w \geq 4.0$ for the instrumental period and extending to smaller magnitudes and earlier periods for historical events. It merges multiple national bulletins (INGV, GNDT, ING), the ISC bulletin, GCMT solutions, and a rich archive of macroseismic intensity data (Locati et al., 2016).

CPTI15 uses a piecewise-linear GOR framework (Gasperini et al., 2012) for magnitude conversion and the inverse-variance-weighted combination of (15) for events with both instrumental and macroseismic estimates. The catalogue is tied to the INGV ShakeMap intensity model to ensure consistency with ground-motion prediction equations used in Italian national hazard maps (Stucchi et al., 2013).

1.8.5 Aotearoa New Zealand

The GeoNet catalog (GeoNet, 2025a) is the authoritative source for New Zealand seismicity, providing real-time and reviewed solutions from the national seismograph network. Illsley-Kemp and Mestel (2025) recently produced a new high-precision catalog for the Taupō Volcanic Zone (TVZ) by re-processing GeoNet waveforms with a template-matching algorithm (Chamberlain et al., 2018), demonstrating that merged catalogs for New Zealand should treat the TVZ and broader crustal seismicity as distinct populations with different completeness thresholds.

The National Seismic Hazard Model (Gerstenberger et al., 2024) integrates the GeoNet catalog with paleo-seismic and geological data, requiring careful synchronisation of magnitude scales: GeoNet reports primarily M_{L} for small events and M_{w} for larger events, but the transition between scales is not uniformly applied across the catalog's recording history (GeoNet, 2025b). For input to ground-motion models, all magnitudes must be expressed in M_{w}, necessitating the conversion procedures discussed in section 1.6.

1.9 Parameter Selection Within a Merged Group

Once event groups have been identified by one of the methods in section 1.3, parameter selection determines the single origin time, hypocentre, and magnitude that will represent the physical earthquake in the merged catalog.

1.9.1 Hypocenter and origin time

The preferred hypocentre is selected from the contributing solutions according to the priority order in table 2, subject to quality thresholds. The ISC-GEM criteria for accepting a prime hypocentre (Storchak et al., 2018; Bondár and Storchak, 2011) are:

- Azimuthal gap $< 180°$,
- At least 6 defining phases,
- Free depth (not fixed at a default value),
- RMS travel-time residual < 2.0 s.

The origin time is taken from the selected hypocentre *without averaging*. Averaging origin times across agencies is incorrect because agencies use different phase sets and velocity models; their origin times are not independent estimates of a common truth but are each internally consistent with their respective location procedures (Storchak et al., 2013).

Where no single agency meets all criteria, the highest-priority solution is used with a downgraded quality flag, and the criteria violation is recorded as event metadata.

1.9.2 Depth selection

Among qualifying hypocentres, the solution with the smallest formal depth uncertainty is preferred. Where formal uncertainty is unavailable, the solution with the largest number of depth-sensitive phases (pP, sP, pwP, depth phases) is preferred. Depth solutions fixed at standard values (10, 33, 100 km) are given lowest priority and flagged with `depth_type = fixed` (Engdahl et al., 2020).

For events with no free-depth solution from any contributing agency, the depth should be reported as the fixed value of the best-available agency, with explicit documentation of the fixing convention used.

1.9.3 Magnitude selection and conversion

Apply the priority hierarchy of section 1.6.3. When multiple estimates exist at the same priority tier and formal uncertainties are available, the Bayesian posterior mean (13) is the optimal estimator (Si et al., 2024). The alpha-trimmed median (Storchak et al., 2018) is an acceptable substitute when the Bayesian framework cannot be calibrated (e.g. for sparse regional catalogs without sufficient overlap for network intercept estimation). When neither is applicable, select the estimate from the highest-priority agency and document the choice.

1.9.4 Focal mechanism

For events where focal mechanisms are available from multiple sources, the following priority order is recommended based on the ISC-GEM protocol and established reliability assessments (Dziewonski et al., 1981; Ekström et al., 2012):

1. GCMT (waveform centroid moment tensor);
2. USGS/NEIC broadband CMT or W-phase solution;
3. GEOFON/GFZ regional moment tensor;
4. GeoNet CMT (for New Zealand events, GeoNet 2025a);
5. INGV CMT (for Italian events);
6. First-motion polarity solution with ≥ 20 station polarities;
7. Automated first-motion solution.

Within each tier, rank by: (i) variance reduction (higher is better), (ii) station polarity count (more is better), (iii) misfit (lower is better). All available mechanisms should be retained as supplementary products to enable downstream comparison studies (Storchak et al., 2020).

1.10 Declustering and Its Relationship to Merging

Declustering — the removal of aftershocks and foreshocks to obtain a Poissonian background catalogue — is a separate post-processing step from merging but is closely related in practice. Two points of interaction are important.

Merging before declustering. Declustering algorithms (e.g. Gardner and Knopoff 1974; Reasenberg 1985; Zaliapin and Ben-Zion 2013) require an internally consistent, homogeneous magnitude scale and a complete event listing to correctly identify cluster members. If two catalogs are merged *after* independent declustering, cluster members that appear in only one source catalog may be lost, biasing the background rate estimate (Hainzl et al., 2016). The correct order is: merge first (retaining all events from all sources), then decluster.

Duplicate detection vs. foreshock-aftershock discrimination. The nearest-neighbour proximity function (section 1.3.3) was introduced specifically to address the case where duplicate detection and aftershock identification are conflated (Tanaka et al., 2022). This is a real operational hazard: aftershocks occurring within seconds of the mainshock are easily misidentified as duplicates by fixed-window methods during the most intense phase of a sequence. Systems that merge and decluster simultaneously (e.g. using a single proximity window) can lose genuine aftershocks and overcount the mainshock, distorting both the b-value and the Omori-law parameters of the sequence (Hainzl et al., 2016; Reasenberg, 1985).

1.11 Data Exchange Standards and Provenance

1.11.1 QuakeML and FDSN

The QuakeML 1.2 schema (ETH Zürich Seismology / QuakeML Working Group, 2015) is the de facto standard for representing earthquake event metadata in XML. It natively supports the representation of multiple competing origin solutions, multiple magnitude estimates per origin, focal mechanism solutions with full moment tensors, and phase arrival data. The `preferredOriginID` and `preferredMagnitudeID` attributes are the schema's mechanism for expressing a selection decision, while retaining all contributing solutions as first-class objects.

FDSN event web services (U.S. Geological Survey, 2025) provide HTTP access to QuakeML records from participating agencies, enabling automated ingestion of source catalogs in a standard format. For merging systems that ingest from multiple FDSN nodes simultaneously, the `contributor` and `author` fields within each QuakeML origin provide the information needed to implement the same-agency rule (section 1.12.2) (ETH Zürich Seismology / QuakeML Working Group, 2015; U.S. Geological Survey, 2025).

1.11.2 Provenance tracking

Any merged catalog intended for research use should preserve the full derivation chain of each event record. The W3C PROV data model (World Wide Web Consortium, 2013) provides a formal language for expressing provenance graphs. For earthquake catalogs, provenance records should document:

- The source catalog identifier and version for each contributing record;
- The association algorithm and threshold values used;
- The selection rule applied for each parameter type (origin, magnitude, focal mechanism);
- The conversion equation and coefficients applied to any magnitude requiring homogenisation, with uncertainty estimates.

The FAIR data principles (Wilkinson et al., 2016) require that merged catalogs be assigned persistent identifiers (DOIs) with version tracking (DataCite, 2025). Where the input catalogs have their own DOIs, the provenance record should cite them explicitly to enable users to trace any merged event back to its source records (GO FAIR Initiative, 2025).

1.12 Conflict Detection and Quality Control

1.12.1 Within-group consistency flags

Quality flags should be computed for every merged event and stored as first-class metadata. Table 3 lists the recommended set, drawing on the ISC bulletin quality model (Bondár and Storchak, 2011; Storchak et al., 2020) and the ANSS ComCat practice (USGS, 2025). Flags are informational unless marked as "Error"; error-severity flags indicate that the group should be dissolved and re-evaluated before the event is included in the merged output.

Table 3: Recommended within-group quality flags for merged earthquake events. Severity: I = informational, W = warning, E = error (group should be dissolved).

Condition	Threshold	Severity	Reference
Azimuthal gap	$> 180°$	W	Bondár and Storchak (2011)
Defining phase count	< 6	W	Bondár and Storchak (2011)
Depth fixed (not free)	any	I	Engdahl et al. (2020)
RMS travel-time residual	> 2.0 s	W	Storchak et al. (2020)
Horizontal uncertainty	> 50 km	W	Bondár et al. (2004)
$\lvert\Delta M_\mathrm{w}\rvert$: two authoritative sources	> 0.3	W	Si et al. (2024)
$\lvert\Delta M_\mathrm{w}\rvert$: converted vs. observed	> 0.5	W	Grünthal et al. (2016)
Magnitude range in group	> 1.0 (for $M < 5$)	W	Tanaka et al. (2022)
Group size (events from different networks)	> 10	E	Benz et al. (2019)
Same network appears twice in group	any	E	Benz et al. (2019)
Spatial epicentre spread	> 100 km for $M < 5$	W	Benz et al. (2019)
Depth range within group	> 50 km (shallow)	W	Storchak et al. (2018)
Wide time spread (informational)	> 30 s for $M < 5$	I	Storchak et al. (2018)

1.12.2 The same-agency rule

If the same network contributes two records to a single merged group, the group almost certainly spans two distinct physical earthquakes. This is because any competent network applies intra-network deduplication upstream of its data submission;

consequently, two records from the same network in the same time- distance window represent genuinely distinct detected events, not two reports of one event (Benz et al., 2019). The group should be dissolved and a secondary pairwise pass performed on its members, constrained by the same-agency rule as in the ILP constraint (8).

1.12.3 Failed-group recovery

When a group fails validation (e.g. because of a same-agency clash or anomalous magnitude range), the correct response is not to emit all members as singletons, but to attempt a constrained re-grouping. A one-level secondary greedy pass within the failed group — where the same-agency rule is enforced as a hard constraint — can recover valid sub-groups (Benz et al., 2019; Tanaka et al., 2022). Sub-groups that still fail validation after the secondary pass are emitted as singletons with a QC flag recording the failed-group origin.

1.12.4 Magnitude completeness audit

The magnitude of completeness M_c should be estimated for the merged catalog in rolling temporal bins (5-year windows are recommended by Woessner and Wiemer 2005) using either the Maximum Curvature (MAXC) or Goodness-of-Fit (GFT) method (Wiemer and Wyss, 2000). Events below M_c must be flagged as potentially incomplete in any bin where M_c is elevated and excluded from b-value or seismicity rate calculations that assume completeness (Aki, 1965). Sudden upward steps in M_c correspond to network deterioration or data gaps and should be documented alongside the network history that explains them (Woessner and Wiemer, 2005).

1.13 Algorithmic Recommendations

Table 4 maps the key algorithmic decisions to the methods recommended by the literature reviewed above, and summarises the current implementation status.

Table 4: Algorithmic decisions in earthquake catalog merging: literature recommendations, current implementation, and priority for future work. GOR = Generalised Orthogonal Regression; OLS = Ordinary Least Squares; DSU = Disjoint-Set Union.

Decision	Recommended	Current	Priority
Event association	Graph DSU (Benz et al., 2019)	Greedy forward-sweep	Medium
Dense sequences	Nearest-neighbour η (Tanaka et al., 2022)	Adaptive windows	Low
Magnitude conversion	GOR / non-linear (Grünthal et al., 2016; Storchak et al., 2018)	Linear OLS (Scordilis, 2006)	**High**

(continued)

Decision	Recommended	Current	Priority
Multi-estimate averaging	α-trimmed median (Storchak et al., 2018) or Bayesian (Si et al., 2024)	Priority selection	Medium
Source provenance	All solutions retained (ISC model)	JSON in `source_events`	Low
Depth selection	Uncertainty-weighted (Engdahl et al., 2020)	Implemented	Done
Focal mechanism	GCMT > regional CMT	Implemented	Done
QC flags	13-flag model (table 3)	9 flags implemented	Low
M_c audit	Rolling 5-yr bins (Wiemer and Wyss, 2000)	Not implemented	Medium
Declustering order	Merge first, then decluster	N/A (user responsibility)	Info
Provenance	W3C PROV records (World Wide Web Consortium, 2013)	Partial (JSON lineage)	Medium

The single highest-priority gap is the regression methodology. Replacing linear OLS Scordilis (2006) coefficients with GOR-derived relationships (Grünthal et al., 2016) would reduce systematic bias in the converted frequency-magnitude distribution without requiring architectural changes. For applications in the New Zealand region specifically, region-calibrated $M_\mathrm{L} \rightarrow M_\mathrm{w}$ GOR coefficients estimated from the GeoNet and ISC-GEM overlap period (1976–present) should be derived and used in preference to global coefficients (GeoNet, 2025b; Gerstenberger et al., 2024).

The second highest-priority gap is the adoption of a Union-Find graph association step (section 1.3.4). The current greedy forward-sweep misses transitive associations that arise when an intermediate-time event B links two events A and C that would not independently satisfy the proximity criterion. A Union-Find graph implementation resolves this in $O(n\,\alpha(n))$ time with no change to the downstream parameter-selection logic.

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